

CUSTOMER BEHAVIOR ANALYSIS

BUSINESS INTELLIGENCE PROJECT

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TABLE OF CONTENTS

- 1 Introduction
- 2 Project Goals and Deliverables
- 3 Implemented Phases
- 4 Project Code
- 5 References

INTRODUCTION

This project aims to analyze customer behavior by extracting, transforming, and loading (ETL) data from various sources into a data warehouse. The goal is to uncover insights on customer trends, preferences, and purchasing habits to guide business decisions. It focuses on demographics, purchase patterns, and seasonal trends to help with customer segmentation and behavior analysis.

PROJECT GOALS AND DELIVERABLES

GOALS:

- TO BUILD A ROBUST DATABASE AND ANALYZE CUSTOMER BEHAVIOR ACROSS DIFFERENT DEMOGRAPHIC AND SEASONAL SEGMENTS.
- TO UTILIZE BUSINESS INTELLIGENCE (BI) TOOLS TO CREATE VISUALIZATIONS, REPORTS, AND DASHBOARDS.

DELIVERABLES:

- A FUNCTIONAL SQLITE DATABASE FOR ANALYSIS.
- INSIGHTS ON CUSTOMER DEMOGRAPHICS, PREFERENCES, AND PURCHASING PATTERNS.
- POWER BI DASHBOARDS AND REPORTS PRESENTING THE INSIGHTS INTERACTIVELY.



MAIN IMPLEMENTED PHASES

1

Collecting And understanding the data

2

Database Design and ETL Process

3

Data Modeling

4

ROLAP and SQL Queries

5

Visualization and Reporting Using Power BI

COLLECTING AND UNDERSTANDING THE DATA

we have collected data about customer behavior
that is split in three files

CUSTOMER_INFO.XLSX
CONTAINS CUSTOMER DEMOGRAPHIC
INFORMATION SUCH AS AGE, GENDER, AND
LOCATION.)

	A	B	C	D
1	Customer	Age	Gender	Location
2	1	55	Male	Kentucky
3	2	19	Male	Maine
4	3	50	Male	Massachusetts
5	4	21	Male	Rhode Island
6	5	45	Male	Oregon
7	6	46	Male	Wyoming
8	7	62	Male	Montana

CUSTOMER_RELATION.JSON
INCLUDES SUBSCRIPTION STATUSES AND
PAST PURCHASES.

```
{
  "Customer ID": "1",
  "Subscription Status": "Yes",
  "Previous Purchases": "14",
  "Frequency OF Purchases": "Fortnightly"
},
{
  "Customer ID": "2",
  "Subscription Status": "Yes",
  "Previous Purchases": "2",
  "Frequency OF Purchases": "Fortnightly"
}
```

PURCHASE_RECORDS.CSV
PROVIDES TRANSACTION-LEVEL DETAILS
LIKE PURCHASE AMOUNTS, PRODUCT
CATEGORIES, SEASONS, AND REVIEW
RATINGS

ID	Item Purchased	Category	Purchase Amount (USD)	Size	Color	Season	Shipping Method	Payment Method	Review Rating	Purchase Date
1	Clothing	53	L	Gray	Winter	3.1	Express	Yes	Yes	Venmo,2024-01-02
2	Clothing	64	L	Maroon	Winter	3.1	Express	Yes	Yes	Cash,2024-02-06
3	Clothing	73	S	Maroon	Spring	3.1	Free Shipping	Yes	Yes	Credit Card,2024-03-10
4	Footwear	90	M	Maroon	Spring	3.5	Next Day Air	Yes	Yes	PayPal,2024-04-15
5	Clothing	49	M	Turquoise	Spring	2.7	Free Shipping	Yes	Yes	PayPal,2024-05-20
6	Footwear	20	M	White	Summer	2.9	Standard	Yes	Yes	Venmo,2024-06-06
7	Clothing	85	M	Gray	Fall	3.2	Free Shipping	Yes	Yes	Cash,2024-11-07
8	Clothing	34	L	Charcoal	Winter	3.2	Free Shipping	Yes	Yes	Credit Card,2024-12-01
9	Footwear	97	L	Silver	Summer	2.6	Express	Yes	Yes	Venmo,2024-06-14
10	Accessories	31	M	Pink	Spring	4.8	2-Day Shipping	Yes	Yes	Cash,2024-07-28

COLUMNS OF OUR INITIAL DATA

- **Customer ID:** A unique identifier assigned to each individual customer
- **Age:** The age of the customer
- **Gender:** The gender identification of the customer,
- **Item Purchased:** The specific product or item selected by the customer during the transaction.
- **Category:** The broad group to which the purchased item belongs.
- **Purchase Amount (USD):** The monetary value of the transaction
- **date:** date of transactions.
- **Location:** The geographical location where the purchase was made
- **Size:** The size specification (if applicable) of the purchased item.
- **Color:** The color variant or choice associated with the purchased item
- **Season:** The seasonal relevance of the purchased item
- **Review Rating:** A numerical assessment provided by the customer regarding their satisfaction with the purchased item.
- **Subscription Status:** Indicates whether the customer is subscribed or not.
- **Shipping Type:** Specifies the method used to deliver the purchased item
- **Discount Applied:** Indicates if any promotional discounts were applied to the purchase
- **Promo Code Used:** Notes whether a promotional code or coupon was utilized during the transaction
- **Previous Purchases:** Provides information on the number or frequency of prior purchases made by the customer
- **Payment Method:** Specifies the mode of payment employed by the customer
- **Frequency of Purchases:** Indicates how often the customer engages in purchasing activities

DATABASE DESIGN AND ETL PROCESS

we Implemented ETL processes to merge and integrate data into SQLite.

EXTRACT

PULLING RAW DATA FROM OUR VARIOUS FILES:

- customer_info.xlsx
- purchase_records.csv
- customer_relation.json

TRANSFORM

- Cleaning, processing, standardizing data, removing irrelevant columns: shipping type and payment method. And creating Important fields: Generation, product_id and date_id.
- **CREATING SEPERATE TABLES TO HOLD DATA:** fact_table, product_dim, customer_dim and date_dim

LOAD

LOADING THE CLEANED AND TRANSFORMED DATA INTO AN SQLITE DATABASE THAT WE CREATED:

customer_behavior.db
FOR ANALYSIS

The code is available in the project code document as well as the Biproject folder

OUR FINAL DATA

FACT_TABLE

customer_id, product_id, date_id, purchase_amount_usd, discount_applied

fact table: stores measurable metrics for customer transactions, making it easy to aggregate and analyze key performance indicators.

CUSTOMER_DIM

customer_id, age, gender, location, subscription_status, previous_purchases, requeryency_of_purchases

PRODUCT_DIM

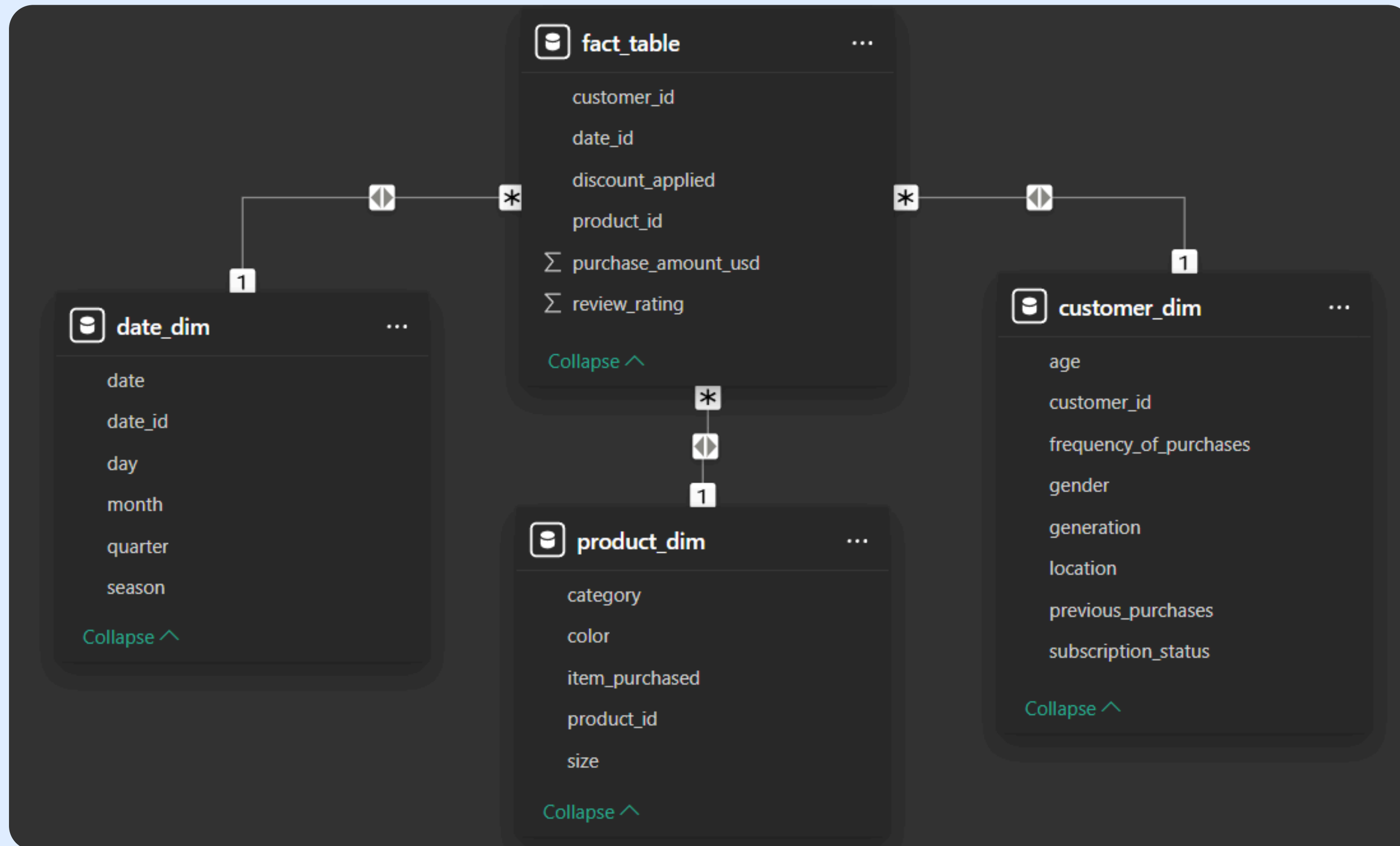
product_id, item_purchased, category, size, color

DATE_DIM

date_id, date, quarter, month, day, season

dimensions tables: provide descriptive attributes that allow for detailed slicing, dicing, and filtering in OLAP operations.

DATA MODELING



The star schema's straightforward relationships (one-to-many) between the central fact table and dimensions enable faster query performance and simplify Power BI visualizations. This structure is ideal for our focus on analyzing trends like purchasing patterns and sales distribution.

ROLAP AND SQL QUERIES

- We used ROLAP (Relational OLAP) for analysis by querying the relational database (SQLite) directly with Python.
- Why ROLAP?:
 - It is supported by SQLite.
 - ROLAP allows us to perform OLAP-style operations (e.g., roll-up, drill-down, slice, and dice) directly on the relational database without precomputing aggregates.
- We performed numerous queries such as: Gender-based purchasing trends, Generational preferences and review patterns, Seasonal purchasing habits, discount impact... A total of 18 queries: The code and results are provided in the project code word document as well as the BIproject folder.

VISUALIZATION AND REPORTING USING POWER BI

We imported the database into Power BI for advanced analysis And Created interactive visualizations and dashboards.



Customer Behavior Analysis: Generational Preferences

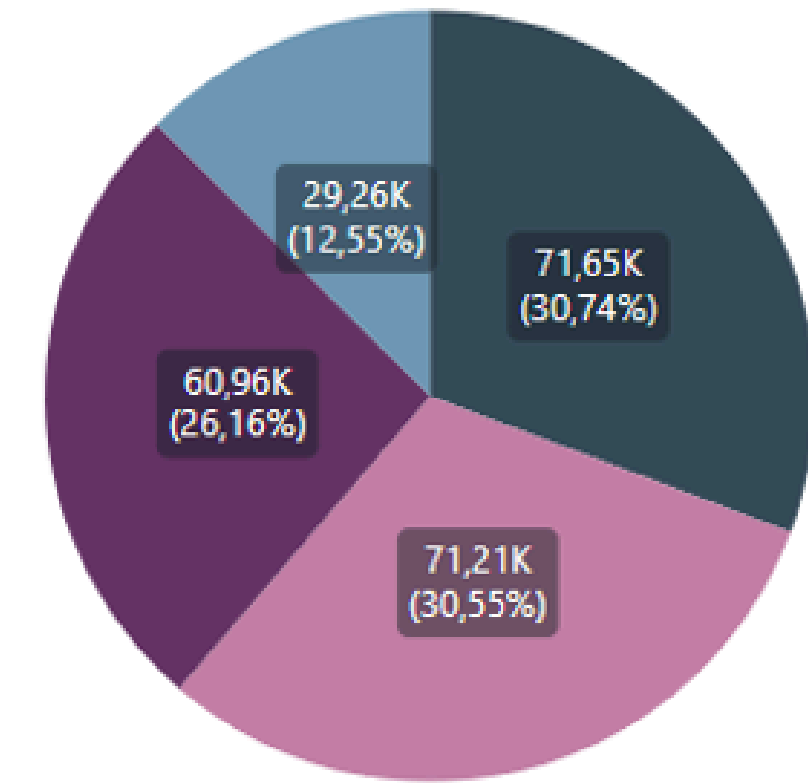
Count of purchase_amount_usd by category and generation

generation ● Baby Boomers ● Gen X ● Gen Z ● Millennials



Sum of purchase_amount_usd by generation

generation ● Gen X ● Millennials ● Baby Boomers ● Gen Z

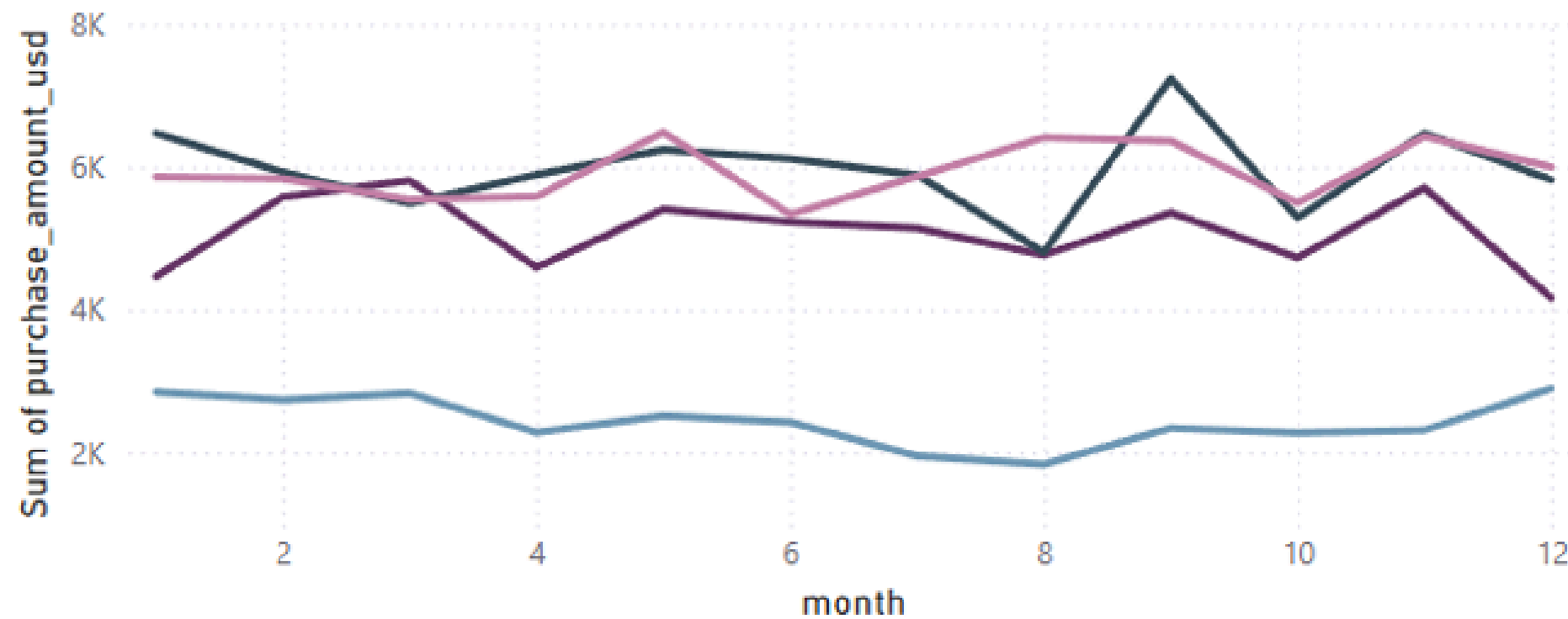


gender
□ Female
□ Male

season
□ Fall
□ Spring
□ Summer
□ Winter

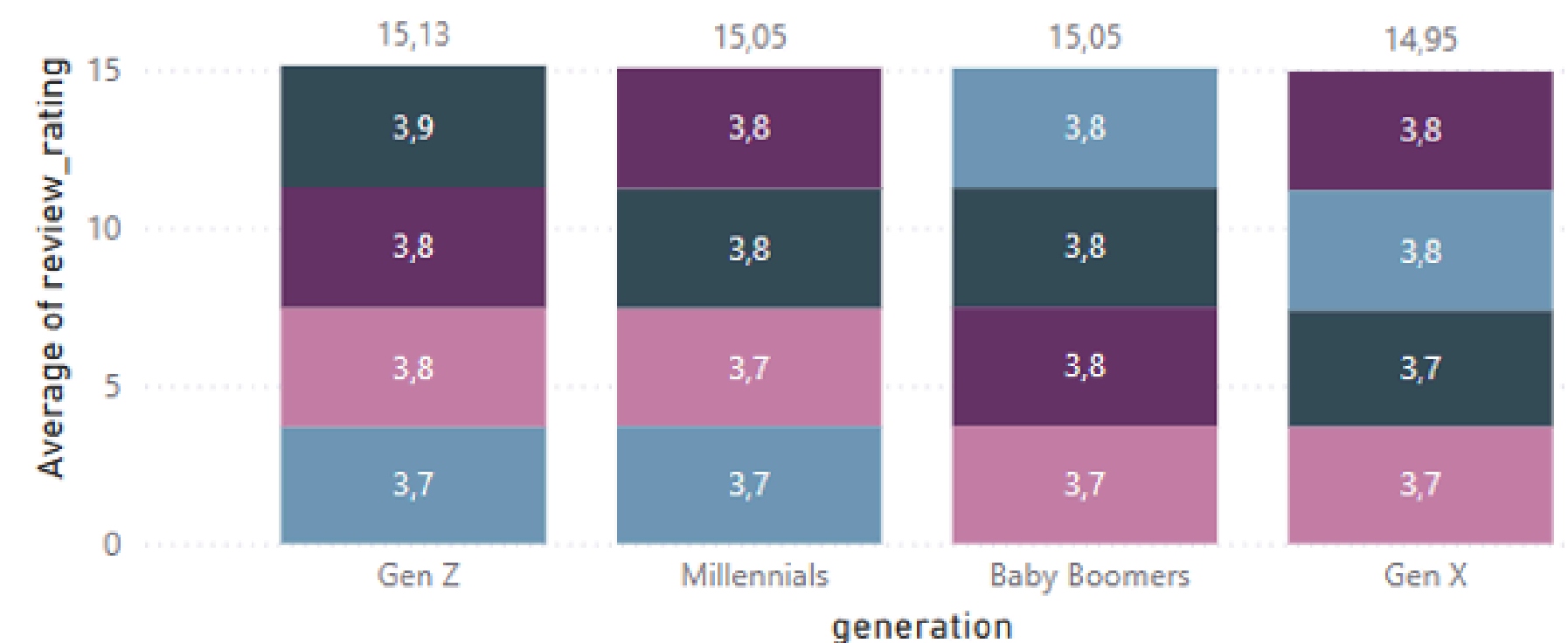
Sum of purchase_amount_usd by month and generation

generation ● Baby Boomers ● Gen X ● Gen Z ● Millennials



Average of review_rating by generation and category

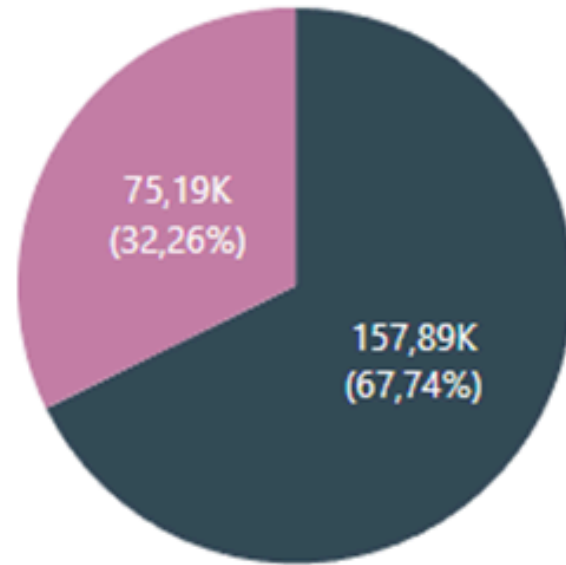
category ● Accessories ● Clothing ● Footwear ● Outerwear



Gender Analysis

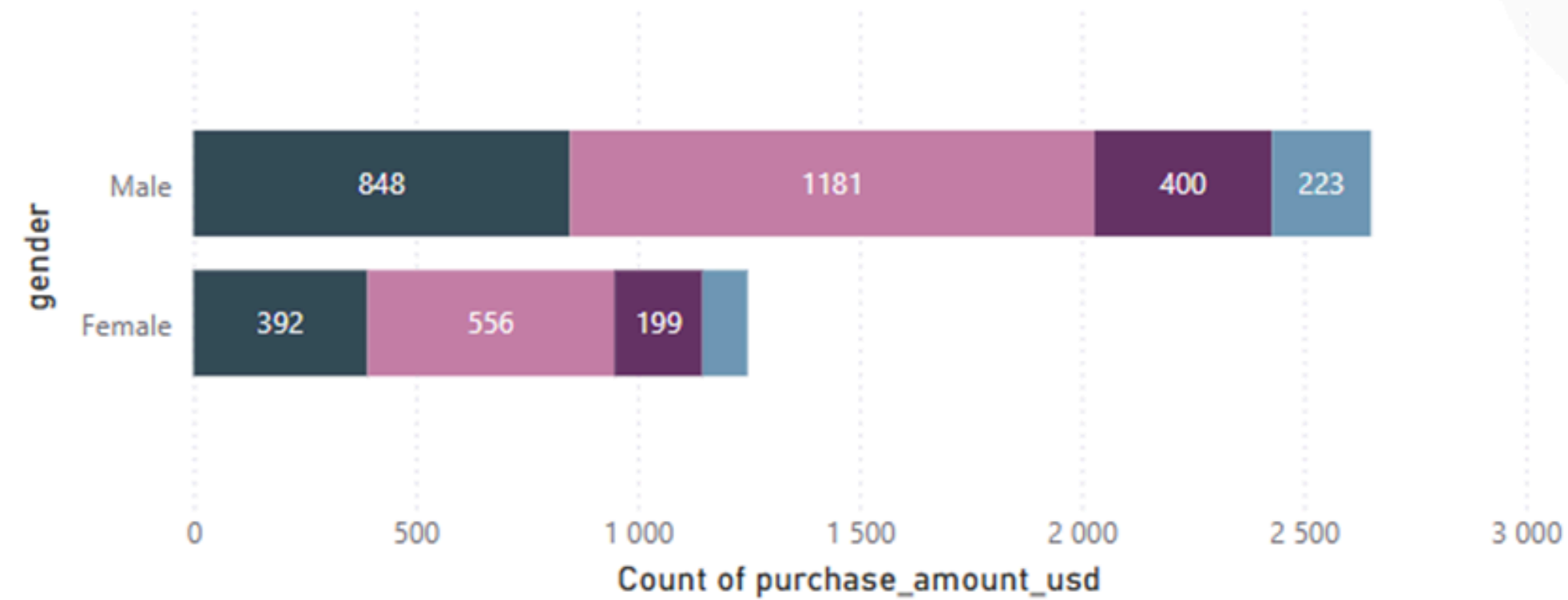
Sum of purchase_amount_usd by gender

gender ● Male ● Female

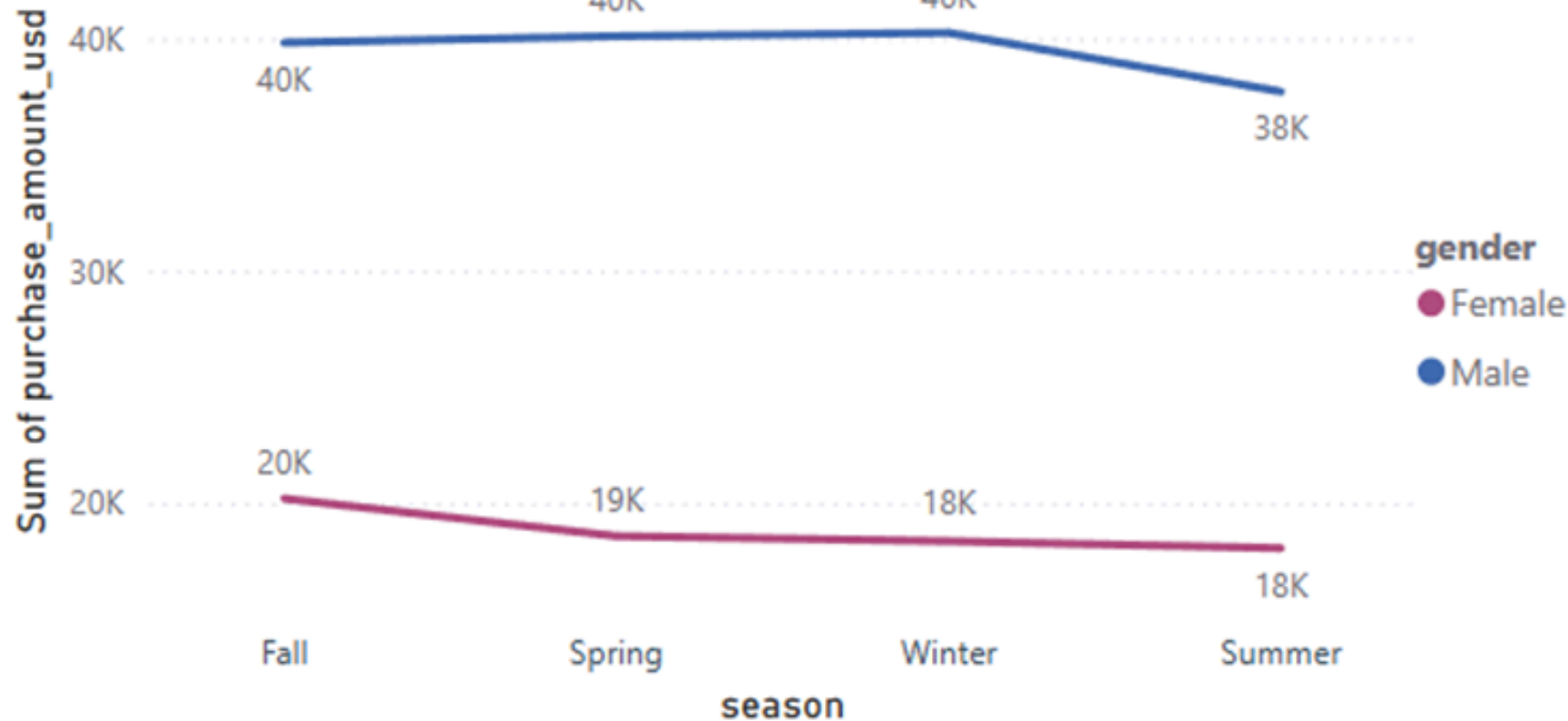


Count of purchase_amount_usd by gender and category

category ● Accessories ● Clothing ● Footwear ● Outerwear

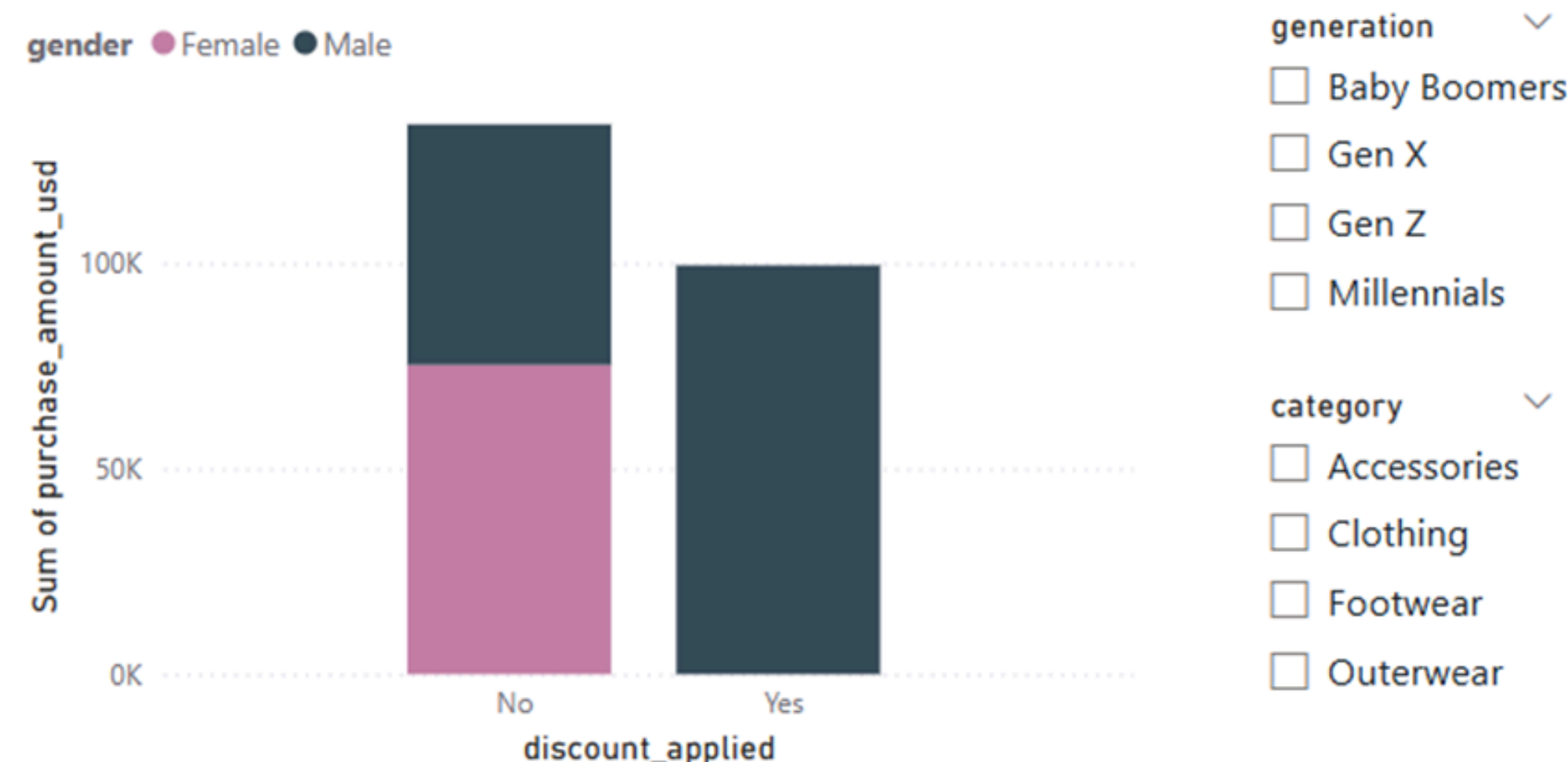


Sum of purchase_amount_usd by season and gender



Sum of purchase_amount_usd by discount_applied and gender

gender ● Female ● Male



generation ▼

- Baby Boomers
- Gen X
- Gen Z
- Millennials

category ▼

- Accessories
- Clothing
- Footwear
- Outerwear

Product Analysis

generation

- ☐ Baby Boomers
- ☐ Gen X
- ☐ Gen Z
- ☐ Millennials

season

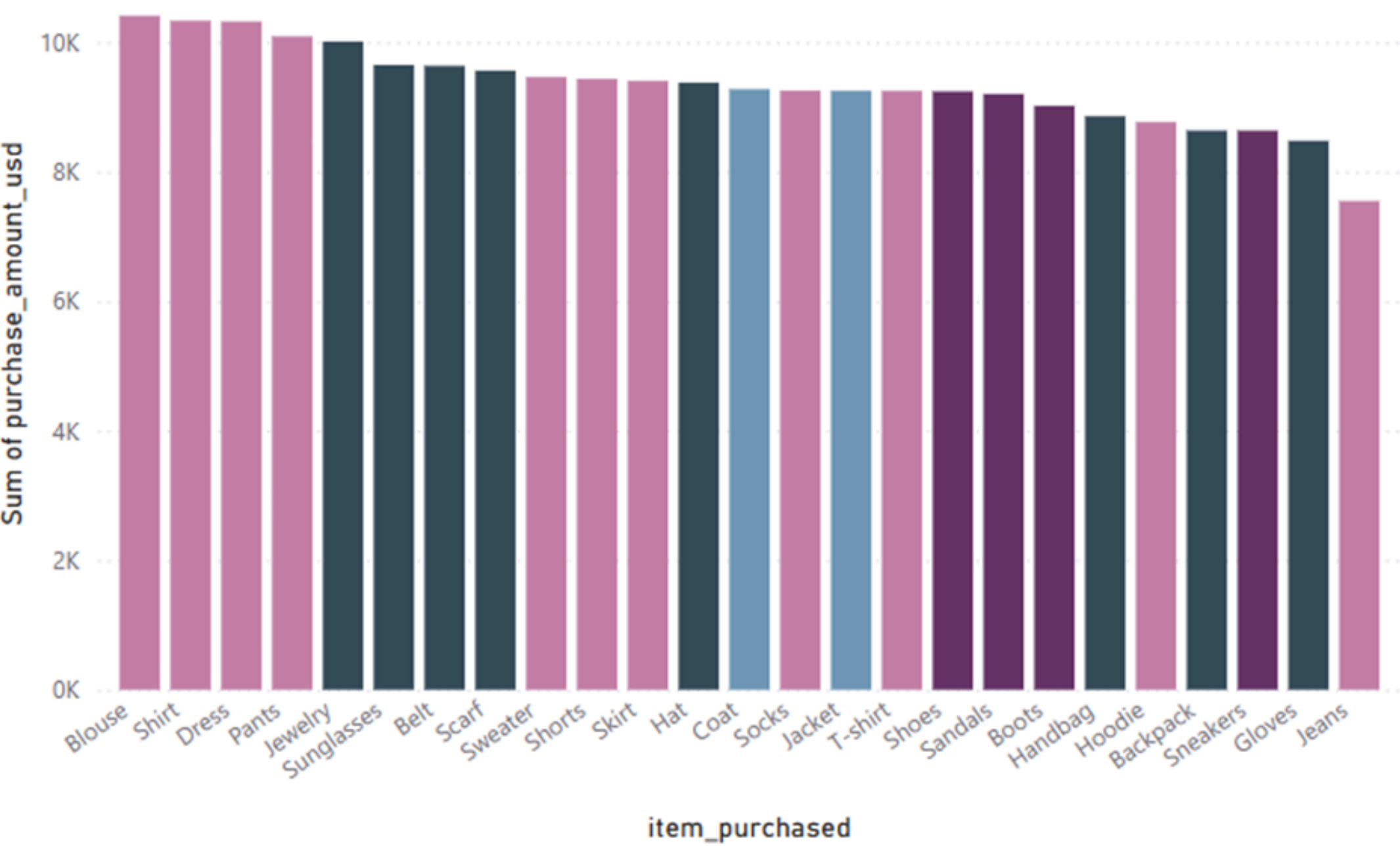
- ☐ Fall
- ☐ Spring
- ☐ Summer
- ☐ Winter

gender

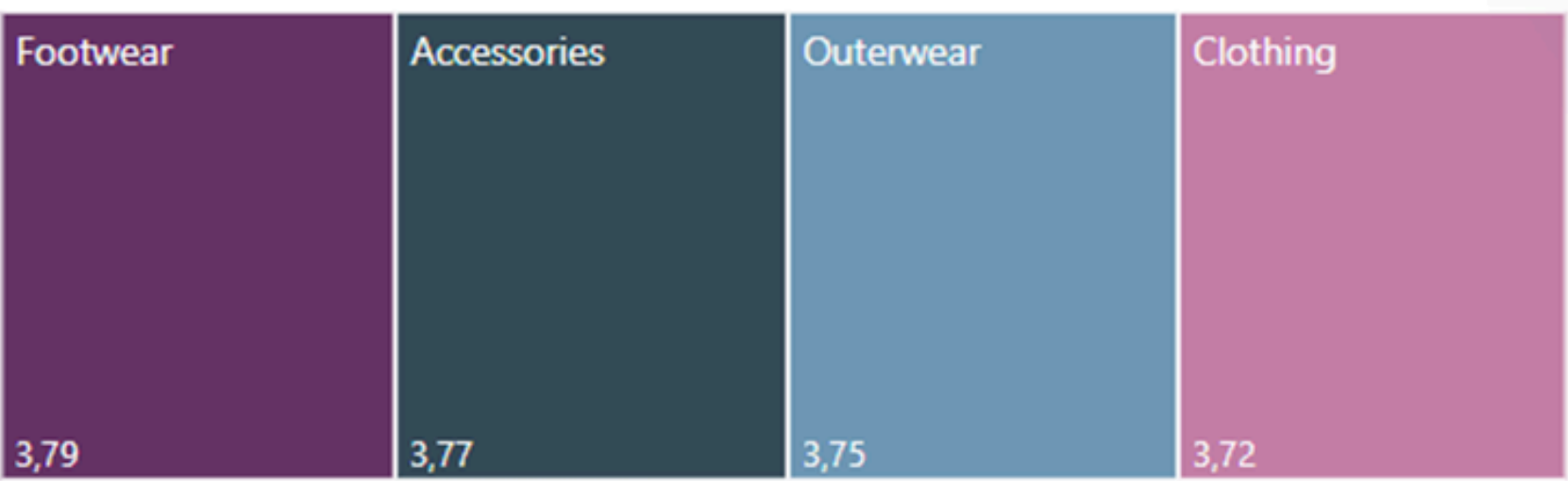
- ☐ Female
- ☐ Male

Sum of purchase_amount_usd by item_purchased and category

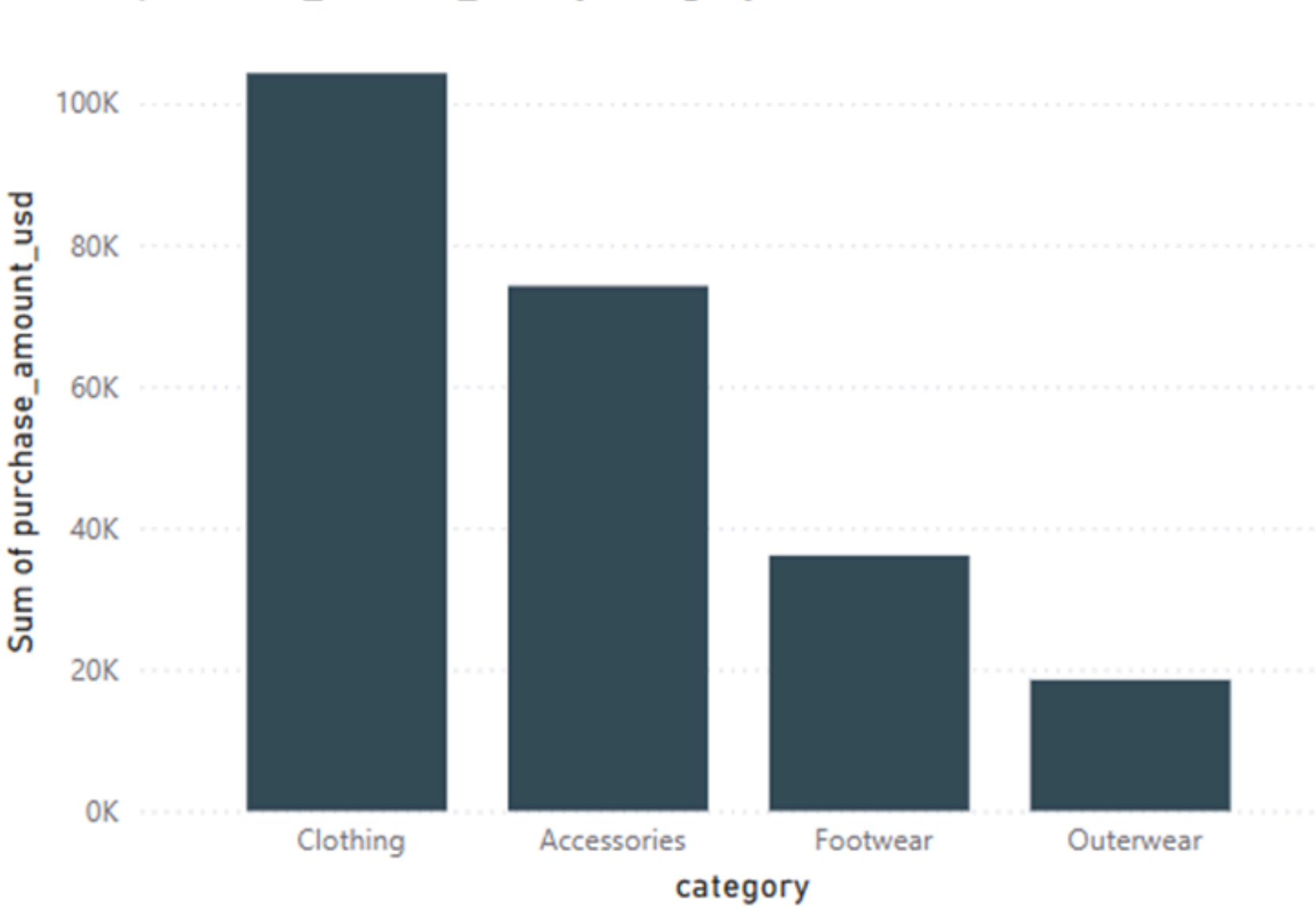
category ● Accessories ● Clothing ● Footwear ● Outerwear



Average of review_rating by category



Sum of purchase_amount_usd by category

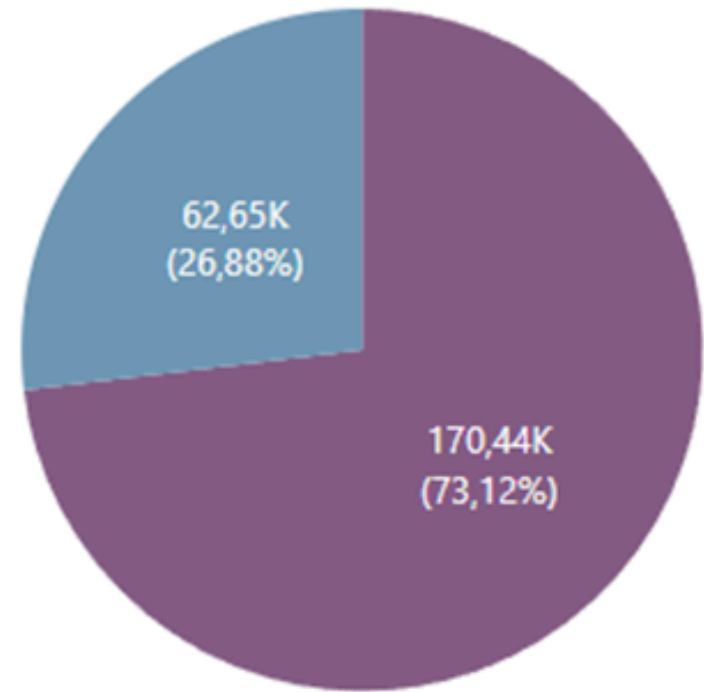


General sales analysis

Sum of purchase_amount_usd by subscription_status

subscription_status

No Yes



Count of purchase_amount_usd by frequency_of_purchases

frequency_of_purchases

Every 3 Months

Annually

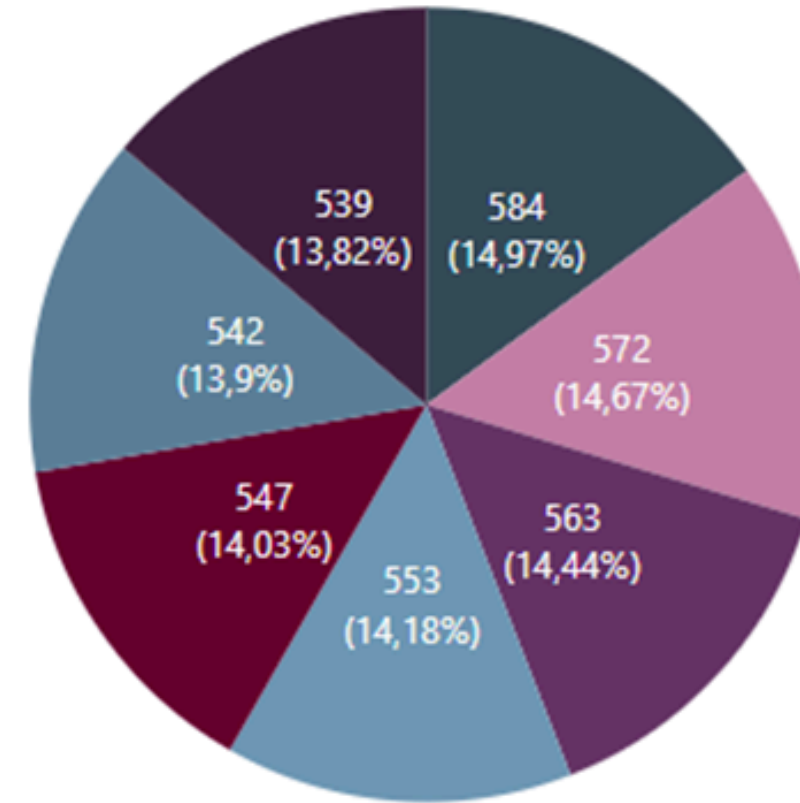
Quarterly

Monthly

Bi-Weekly

Fortnightly

Weekly



Sum of purchase_amount_usd by season

Sum of purchase_amount_usd

60K

58K

56K

Fall

Spring

Winter

Summer

season

Count of purchase_amount_usd and Average of review_rating by location



Summary Analysis

TOTAL PURCHASES

233,08K

Sum of purchase_amount_usd

AVERAGE OF REVIEWS

3,75

Average of review_rating

MAXIMUM AMOUNT SPENT ON PURCHASES

100,00

Max of purchase_amount_usd

NUMBER OF CUSTOMERS

3900

Count of customer_id

NUMBER OF PRODUCTS

25

Count of item_purchased

AVERAGE AMOUNT SPENT ON PURCHASES

59,76

Average of purchase_amount_usd

MAXIMUM REVIEW

5,00

Max of review_rating

MINIMUM REVIEW

2,50

Min of review_rating

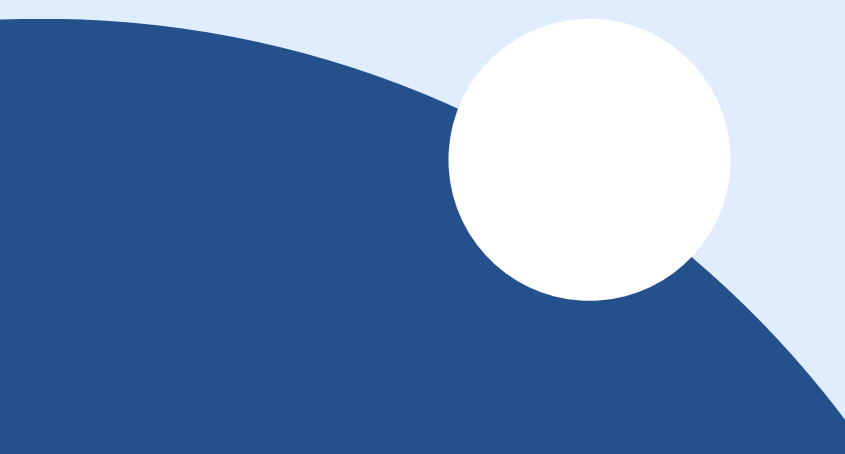
INSIGHTS DERIVED FROM THE VISUALIZATIONS

- The data indicates that younger customers are the most active buyers, but older customers tend to spend more per transaction.
- Seasonality plays a big role, with significant spikes in purchases during holiday periods.
- Clothing and accessories dominate sales, likely due to frequent promotions in these categories.
- Discounts drive higher sales volumes, confirming the importance of promotional strategies.
- Geographic trends show a strong presence in urban centers, pointing to possible opportunities for expansion in underserved regions.

REFERENCES

The project leverages **Python** for data manipulation and database interaction, employing libraries such as **Pandas** and **SQLite**. It also uses **Power BI** for creating interactive dashboards and reports to visualize the findings.

Refer for our BIproject folder and the project code word document for more details



**THANK YOU
FOR YOUR
ATTENTION**